VICTORY 2027

Soil Resources

LECTURE-1 Geography

Sameer Sir



Topics To Be Covered

1. Types of soils
2. **Alluvial Soil || Black and Red Soil**
3.
4.

Soil Resources

- *Types of soil (alluvial, black, red and laterite), mode of formation, distribution, composition and characteristics such as colour, texture, minerals and crops associated.*
- *Soil Erosion – meaning, agents and causes.*
- *Soil Conservation – importance and methods*

Captain Sameer Sir



Hello Avengers

Telegram

https://t.me/icsebysameersir



Soil → A thin layer of small rock particles mixed with organic material that supports plant life is known as Soil.

Humus Content → Decayed organic material
- → Dead Plants
- → Dead Animals

Formation of soil
- → i) Breaking of rocks into small particles.
  - → weathering
  - → Decomposition by Earthworms.
- → ii) Mixing of Humus Content.

Weathering → Breaking down of rocks due to the action of Weather.

Parent rock → Rock from which soil is formed.



Soil Resources

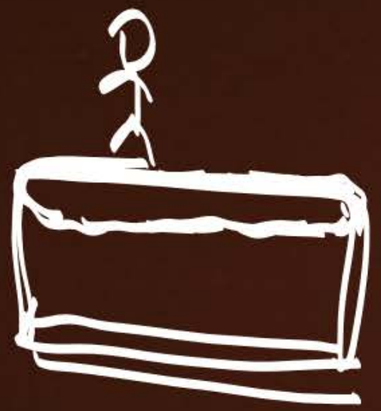
- **Soil is the thin layer of loose mixture of small rock particles and rotting organic matter that covers the surface of the earth.**
- It supports forests, grasslands and crops from which all living creatures on earth derive their energy.



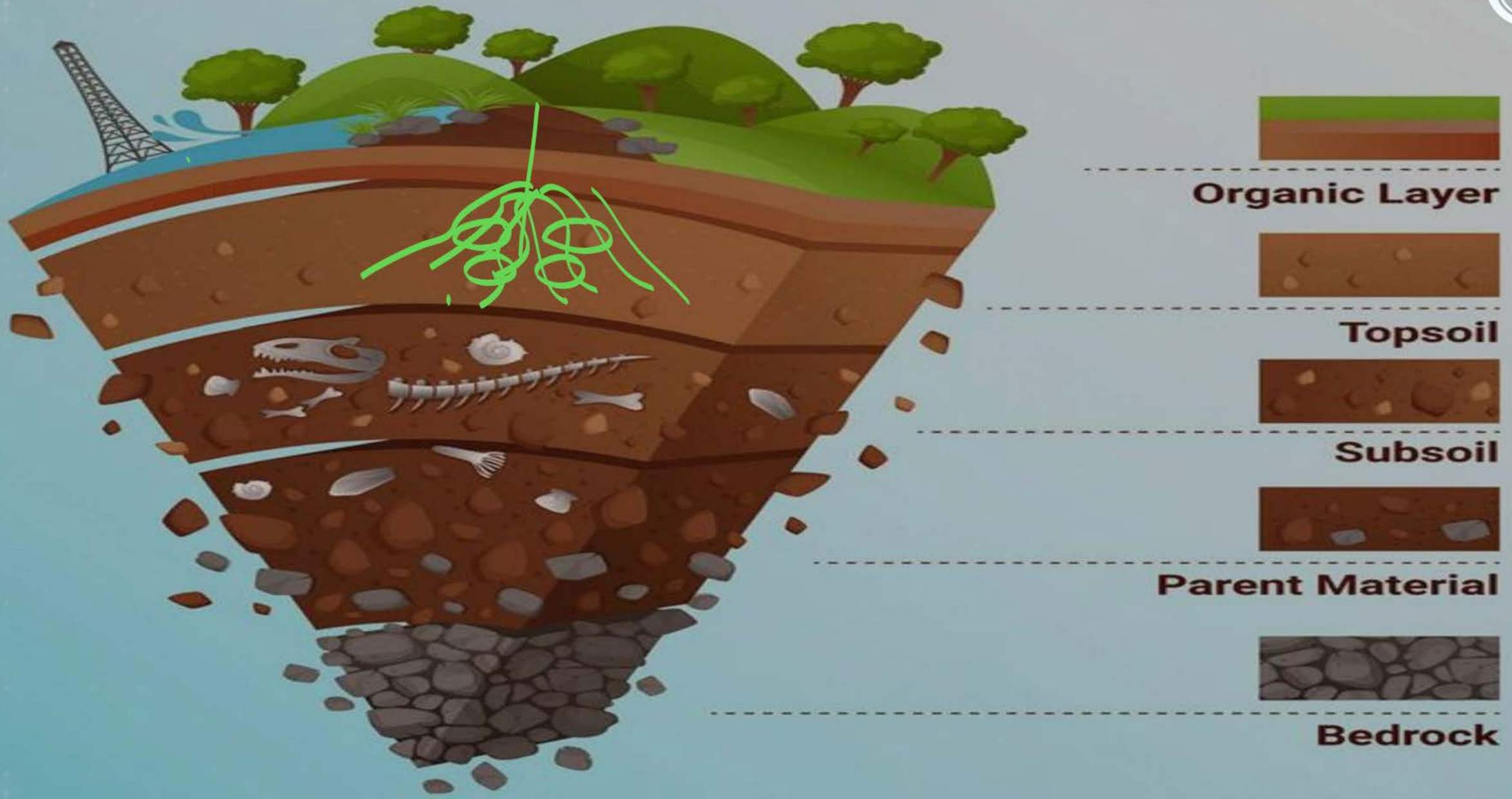
Soil Resources

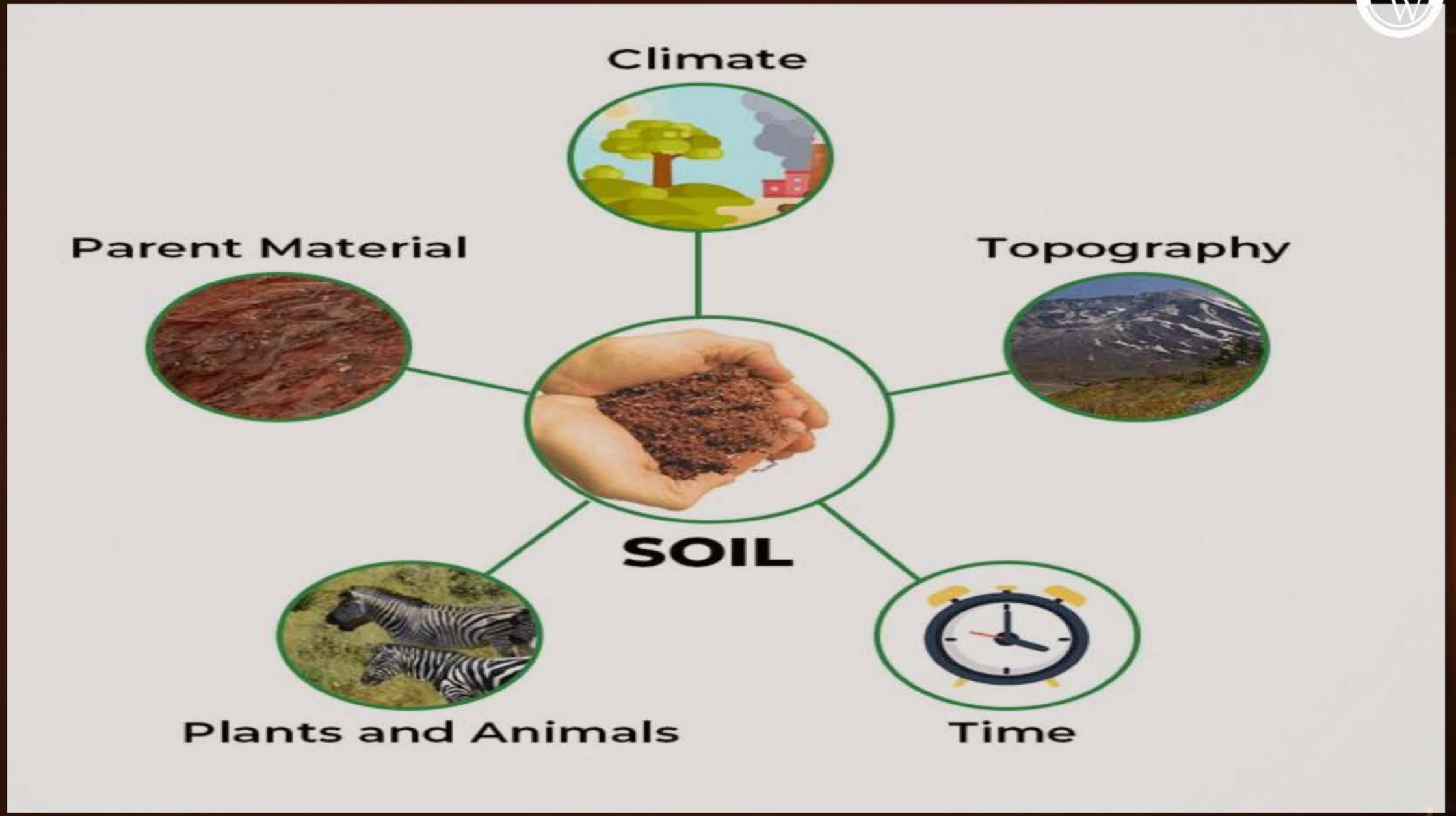
- Various forces of nature such as changing temperature, running water and wind affect formation of soil.
- These forces of nature along with the chemical and organic changes that take place in the soil contribute to the evolution of soil.
- Thus soil itself, has evolved over millions of years.

Soil Resources



- **Faulty agricultural practices, overgrazing and deforestation lead to soil erosion.**
- Improved agricultural practices, reduced pressure of grazing and afforestation can help in the conservation of soil.


Organic Layer
Topsoil
Subsoil
Parent Material
Bedrock


Climate
Parent Material
Topography
SOIL
Plants and Animals
Time

Constituents of Soil

- Soils are derived from parent rock material through a process of breakdown or wear and tear.
- Decomposed vegetal and animal remains, called humus, is an important constituent of soil and adds to the fertility of the soil.
- Besides humus, silica, clay and sand are the other constituents of soil.

Soil Classification

In-situ - on site
Ex-Site → Other site

According to its location, soil can be categorised as:

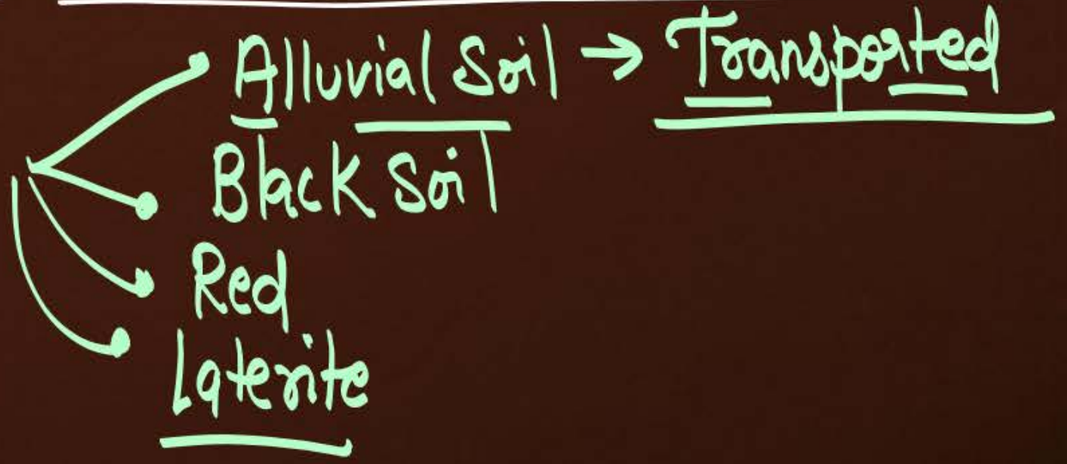


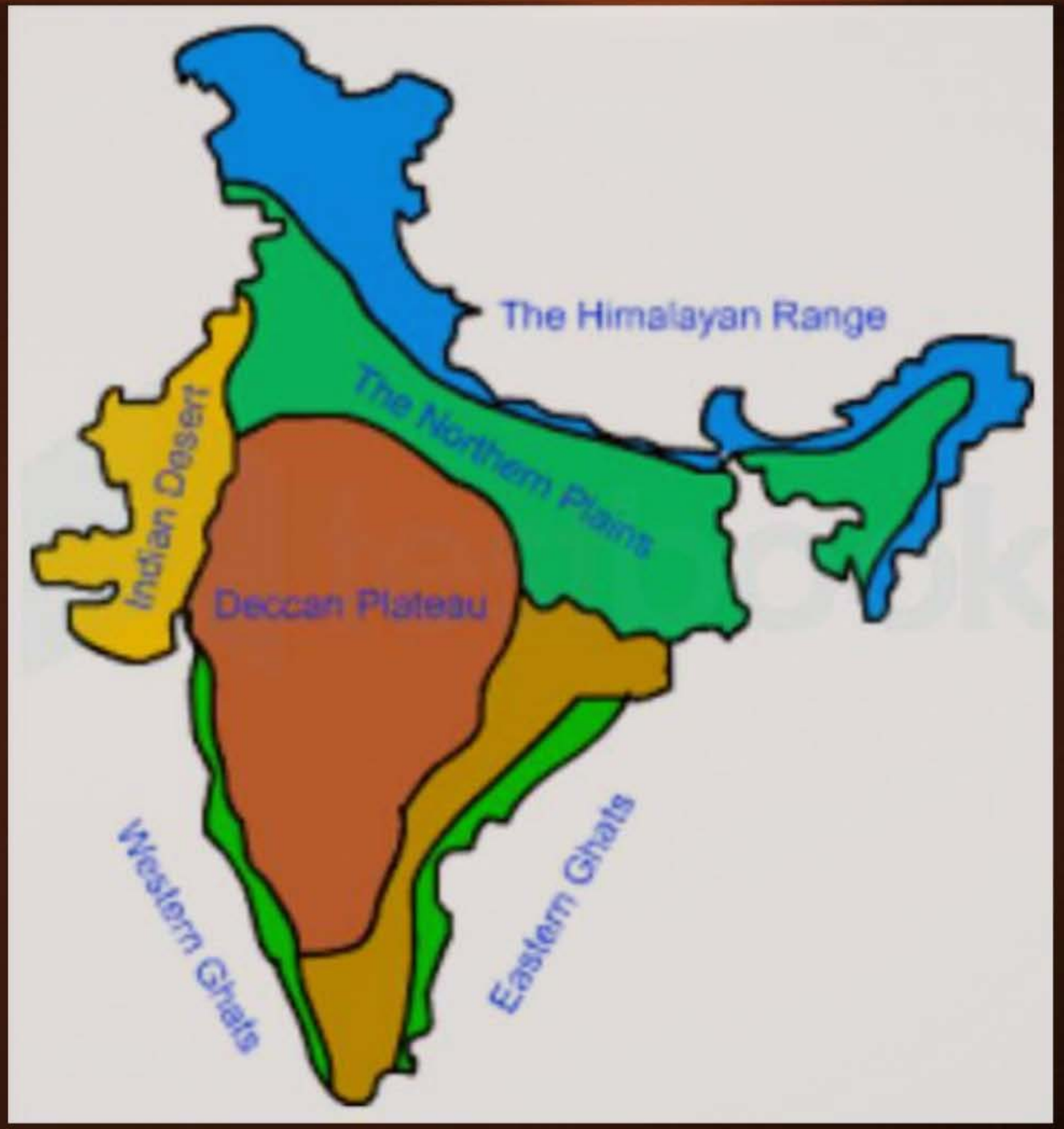
(i) **Residual or Sedimentary Soil:** Almost all soils are formed 'in situ', that is, they are formed in their original position by the breaking down of parent rocks. They have a well defined soil profile. They include black soil, red soil, laterite soil, desert soil, etc.

(ii) **Transported Soil:** These soils are 'ex situ', that is, transported by various agents of erosion and consist of sediments carried and deposited by rivers and winds. Hence, they do not have a well defined soil profile. Alluvial soils are transported soils.

TYPES OF SOIL

- India is a vast region with varied natural environment.
- We find several physiographic regions and different types of climate.
- Therefore, a number of soil types have developed here.
- On the basis of their origin, colour, composition and location the soil of India have been classified into the following types.

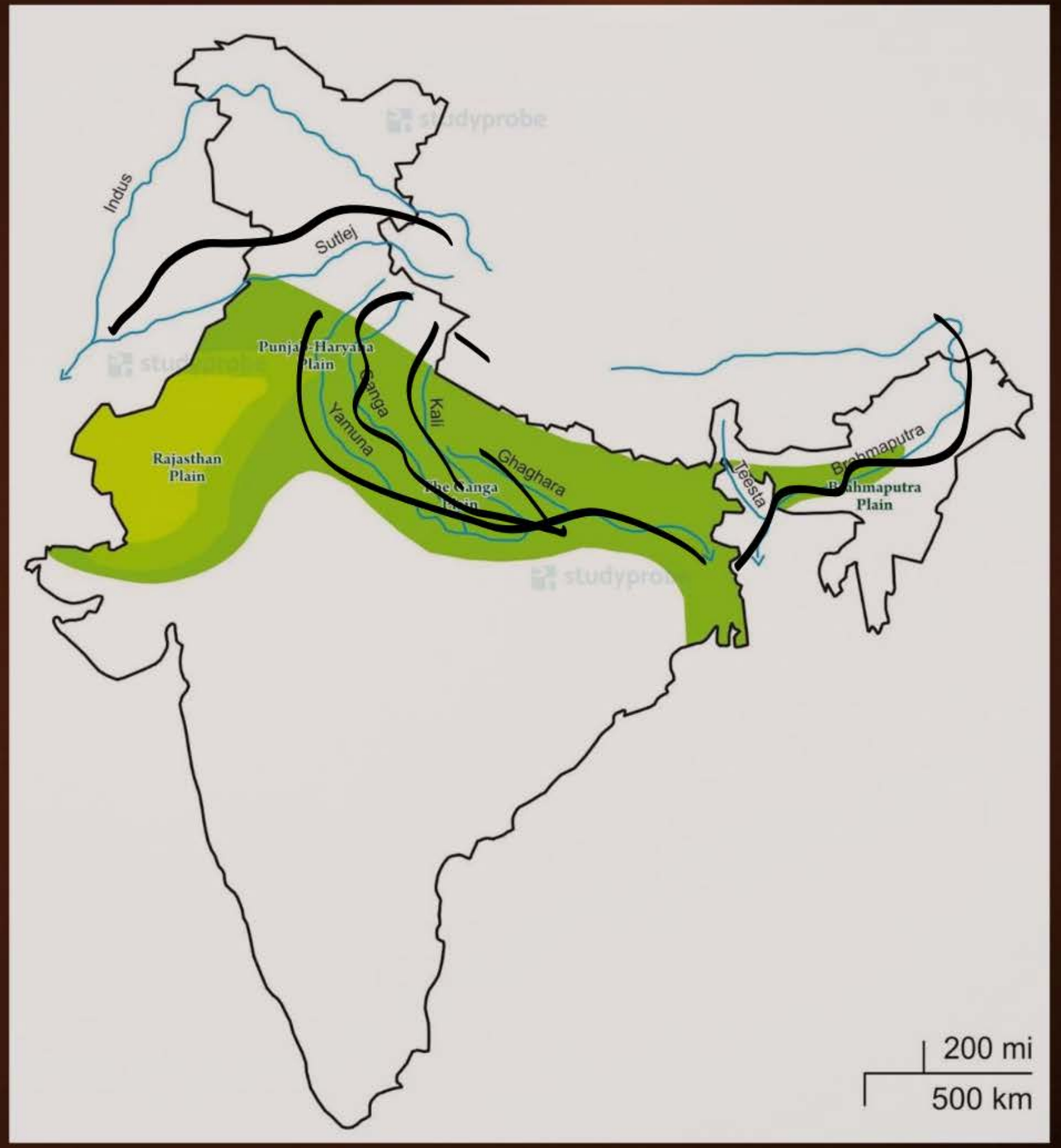



The Himalayan Range
The Northern Plains
Indian Desert
Deccan Plateau
Western Ghats
Eastern Ghats


INDIA
MAJOR SOIL TYPES
68°E
72°
76°
80°
84°
88°
92°
96°E
36°N
32°
28°
24°
20°
16°
12°
8°N
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
Tropic of Cancer
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
LAKSHADWEEP (INDIA)
SRI LANKA
ANDAMAN & NICOBAR ISLANDS (INDIA)
Forest & Mountainous
Alluvial
Red and Yellow
Black
Laterite
Arid
0 200 400 600 km



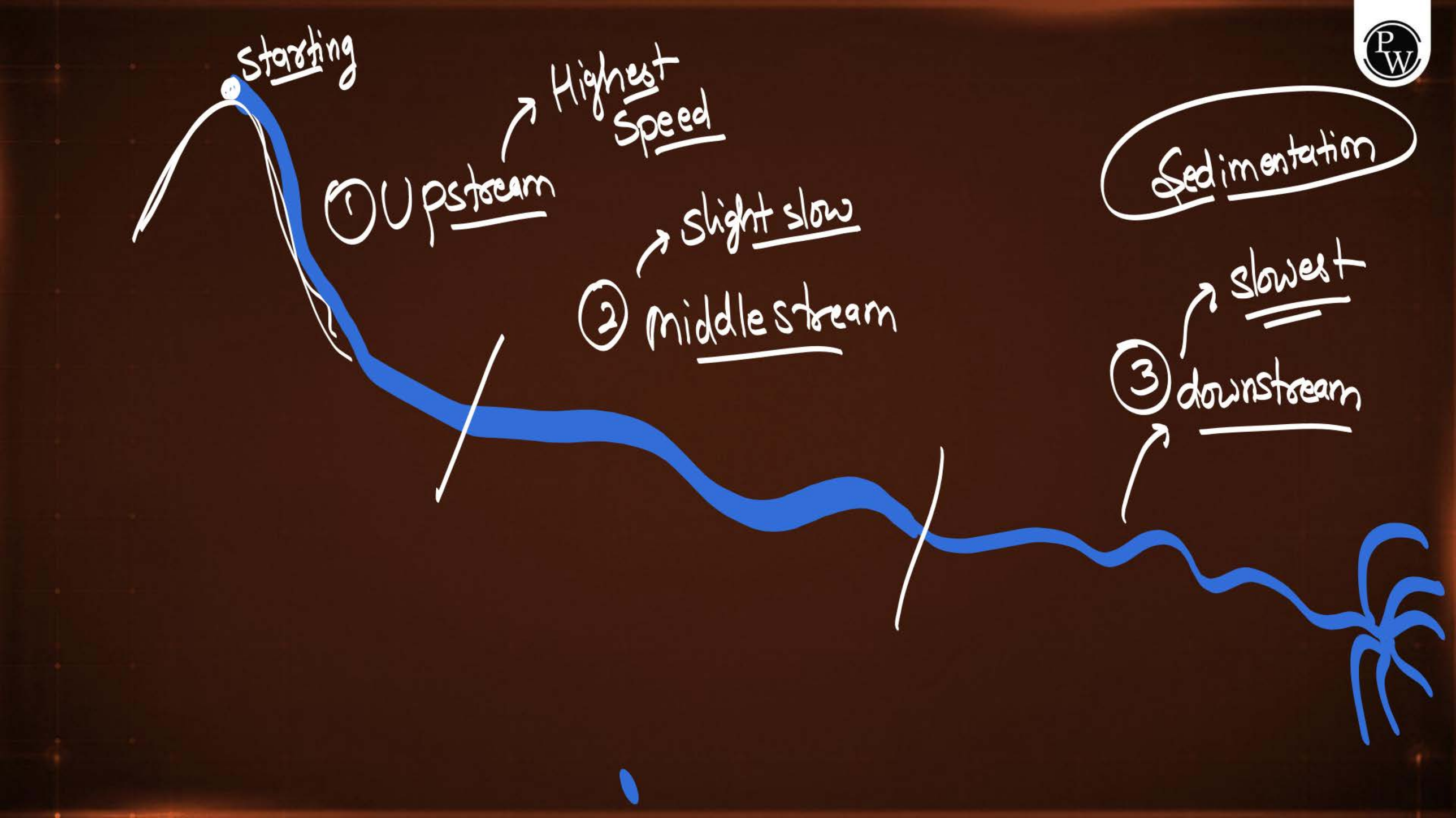
Alluvial Soil


Indus
Sutlej
Punjab-Haryana Plain
Ganga
Yamuna
Kali
Ghaghara
Rajasthan Plain
Teesta
Brahmaputra
Brahmaputra Plain
200 mi
500 km

1. Alluvial Soil

Collection of small particles

- This soil is formed by the sediments brought down by rivers.
- It is rich in chemical ingredients.
- The rivers deposit very fine particles of soil called alluvium in their plains during the course of their long journey.
- The resultant alluvial soil is also known as riverine soil because it is mainly found in the river basins.
- It is a mixture of sand, clay and silt, called loam.


Starting
① Upstream
Highest Speed
② middle stream
slight slow
③ downstream
slowest
Sedimentation

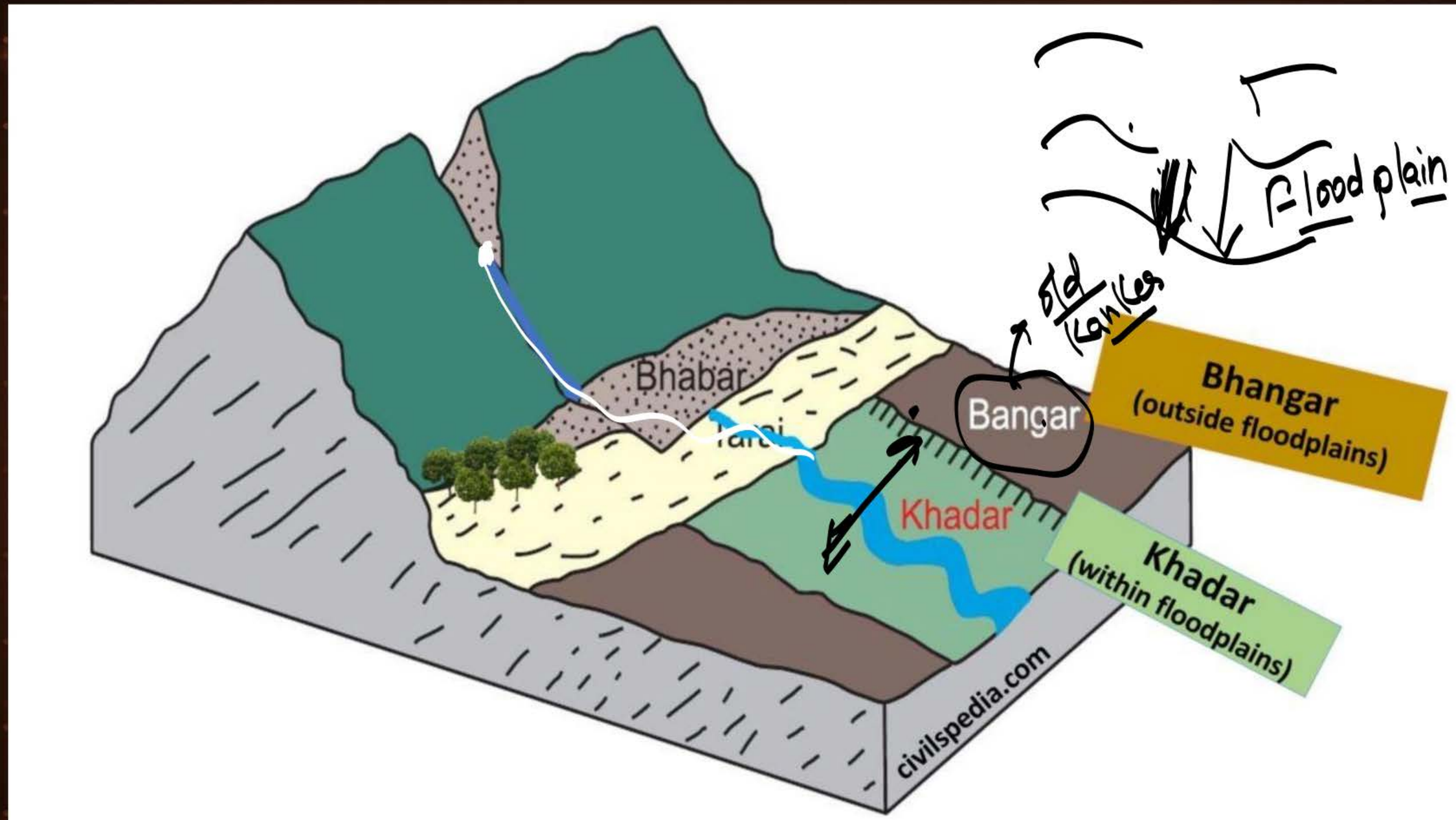
1. Alluvial Soil

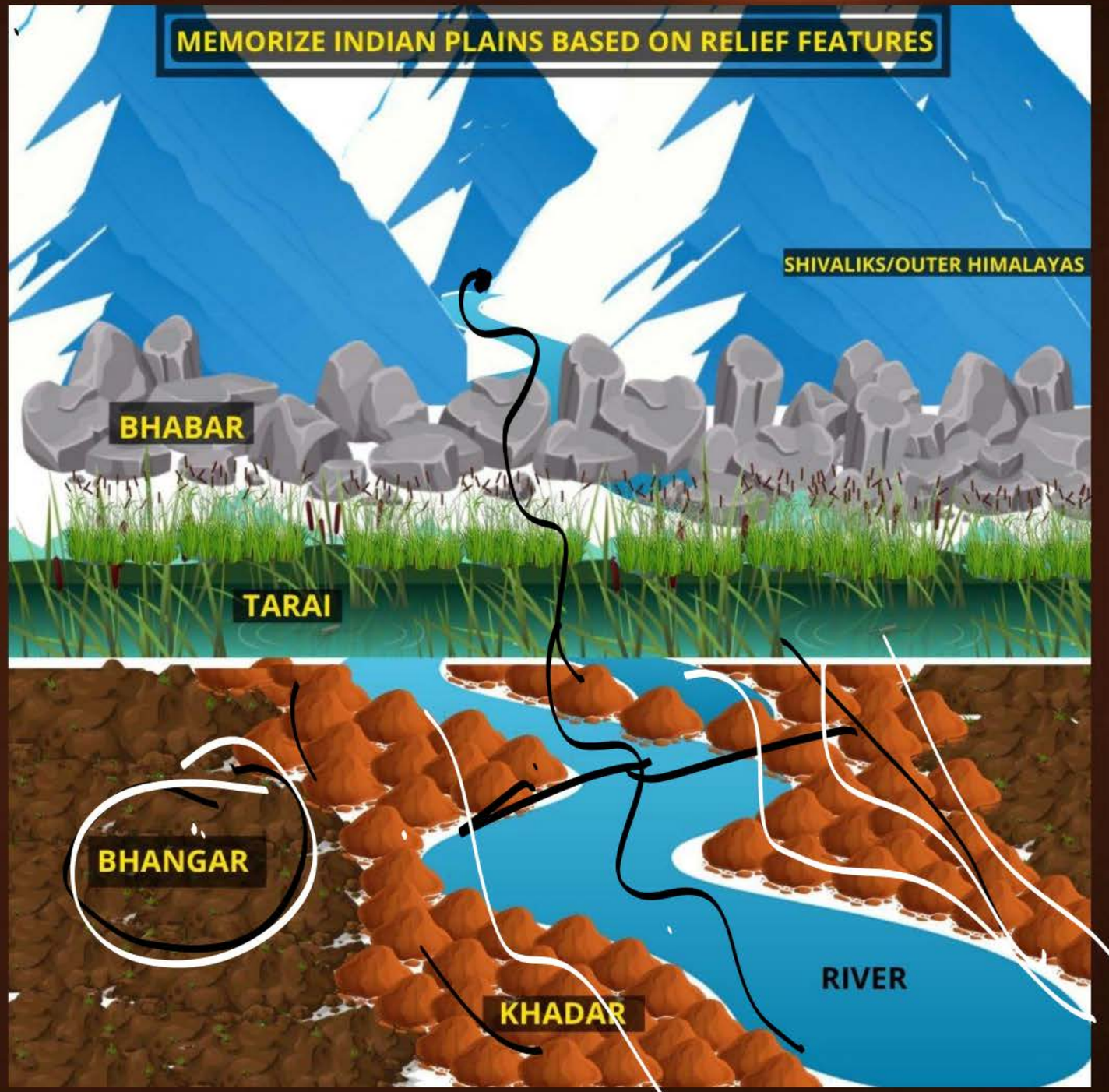


- In India alluvial soils have been deposited by three important river systems - the Indus, the Ganga and the Brahmaputra.
- This soil occupies 40 per cent of the land area.
- The entire Northern Plains are made up of this soil.
- It is also predominant in coastal plains and deltas particularly in the deltas of the Mahanadi, the Godavari, the Krishna and the Kaveri rivers.

1. Alluvial Soil

- On the basis of its age, the alluvial soil is of two types- bhangar and khadar.
- The bhangar or the older alluvium is composed of lime nodules or kanker and has a clayey composition.
- The khadar is light in colour and is composed of newer deposits.


Flood plain
Old Alluvium
Bhabar
Bangar
Bhangar
(outside floodplains)
Khadar
Khadar
(within floodplains)
civilspedia.com


MEMORIZE INDIAN PLAINS BASED ON RELIEF FEATURES
SHIVALIKS/OUTER HIMALAYAS
BHABAR
TARAI
BHANGAR
KHADAR
RIVER

On the basis of age → Bhangar, Khadar

Bhangar Soil	Khadar Soil
➢ Old alluvium soils, known as Bhangar, occupy the largest part of northern plains.	➢ The area in northern plains that lies below flood levels is made up of Khadar.
➢ Bhangar is found above the flood levels of rivers and presents a terrace like structure.	➢ Khadar is younger alluvium as flood waters deposit new layers every year.
➢ Bhangar is full of kankars (lime nodules).	➢ It is formed of fine silt and clay.
➢ Bhangar is not very fertile.	➢ Khadar is very fertile and ideal for intensive cultivation.

1. Alluvial Soil

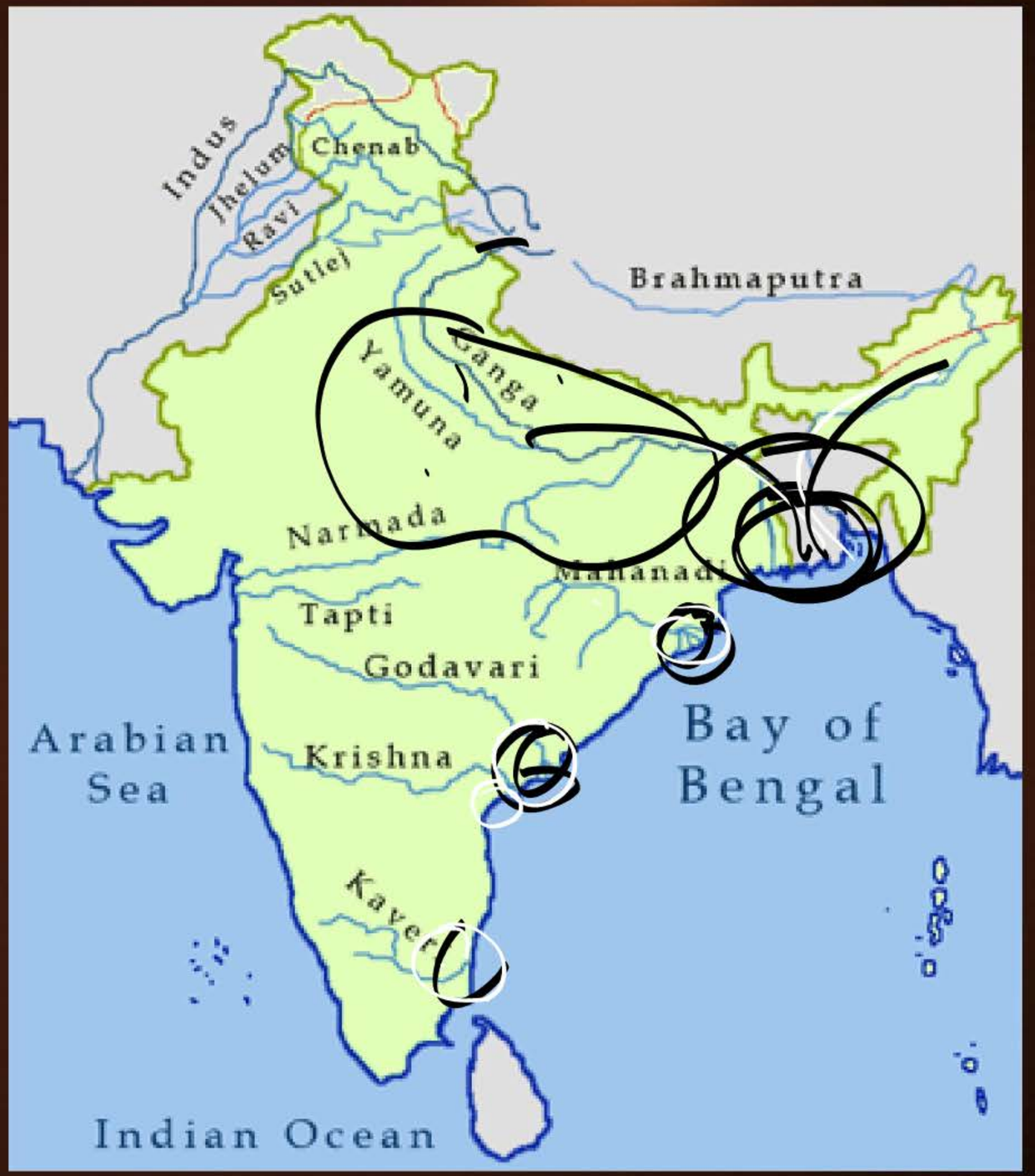
- Khadar is more fertile than bhangar soil as new layers are deposited year after year during monsoon floods.
- Crops can be grown on old alluvial soil by using manure.
- It forms the largest and most important group as it contributes the largest share to the agricultural wealth of India.

1. Alluvial Soil

Distribution

I. **Inland Alluvium:** It is found on the plains of the Indus, the Ganga and the Brahmaputra rivers, extending from Punjab and Sindh (Pakistan) to Bangladesh and Assam (India).

- Punjab, Haryana, Uttarakhand, Uttar Pradesh, Bihar, West Bengal have alluvial soil.
- This soil type also covers a part of Gujarat and a few patches in Rajasthan.


Indus
Jhelum
Chenab
Ravi
Sutlej
Brahmaputra
Yamuna
Ganga
Narmada
Mahanadi
Tapti
Godavari
Krishna
Arabian Sea
Bay of Bengal
Indian Ocean

1. Alluvial Soil

II. **Deltaic Alluvium:** The Deltaic alluvium is found in the deltas of the Ganga-Brahmaputra, Mahanadi, Godavari, Krishna and Kaveri rivers.

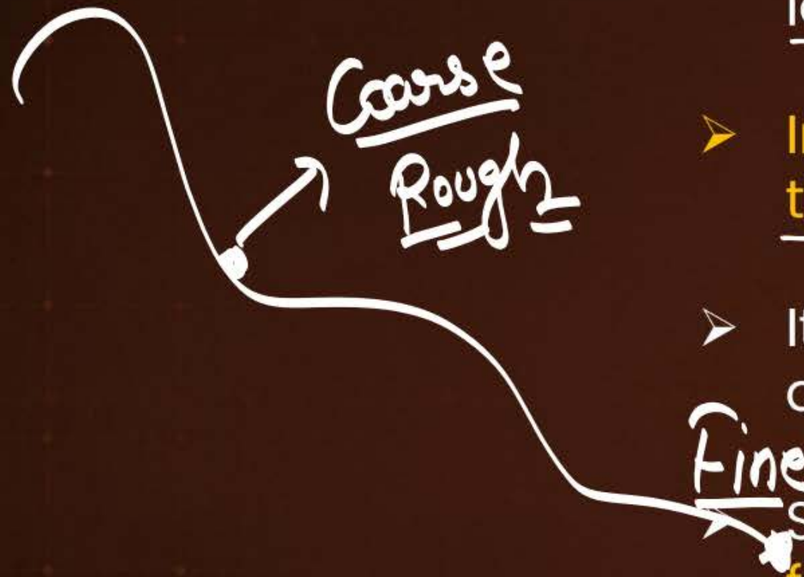
III. **Coastal Alluvium:** It is of tidal origin. It is found in the coastal strips of Peninsular India. It is also found in the plains of Gujarat.

1. Alluvial Soil

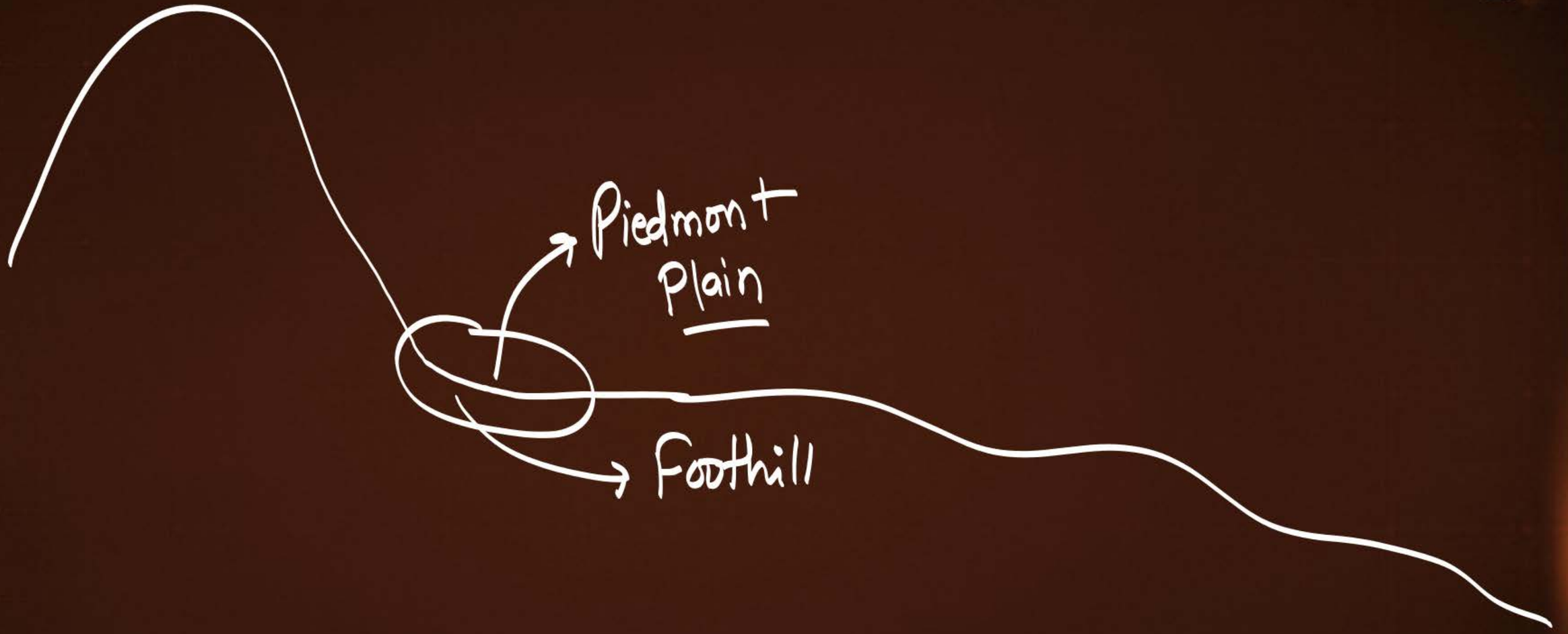
Characteristics of Alluvial Soil

- **Colour:** The colour of the alluvial soil varies from light grey to deep black. Its shades depend on the depth of deposition, the texture of the materials and the time taken for maturity.

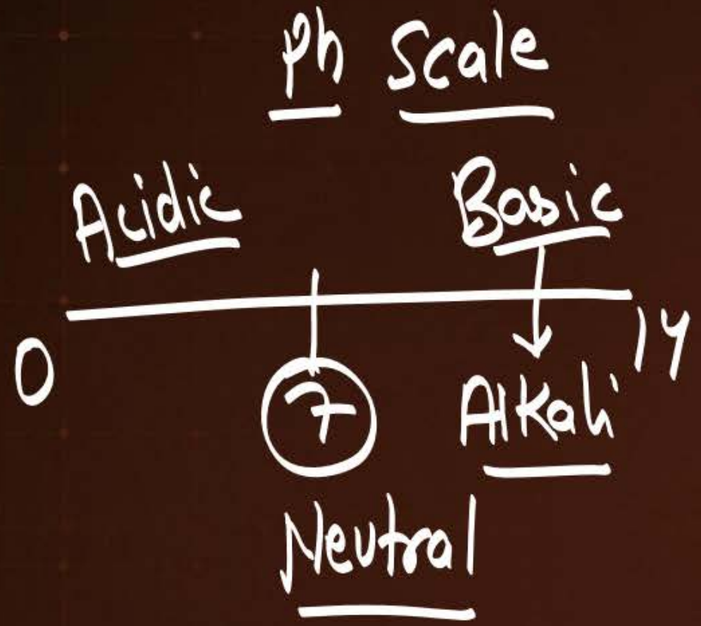
1. Alluvial Soil



- **Texture:** Alluvial soil is formed by the deposition of the river load as it flows from its upper to its lower course.
- In the upper reaches of the river valleys (near the place of their origin) the soil is dry, coarse, porous and sandy.
- It consists of clay and organic matter. Soil particles are large and non-uniform.
- Such soil is predominant in piedmont plains (plains near the foot of mountains).
- As we move further down a river valley, the soil particles become smaller and more uniform.
- They are more compact, less coarse and more moist. They are found up to a depth of 500 metres.


Piedmont
Plain
Foothill

1. Alluvial Soil



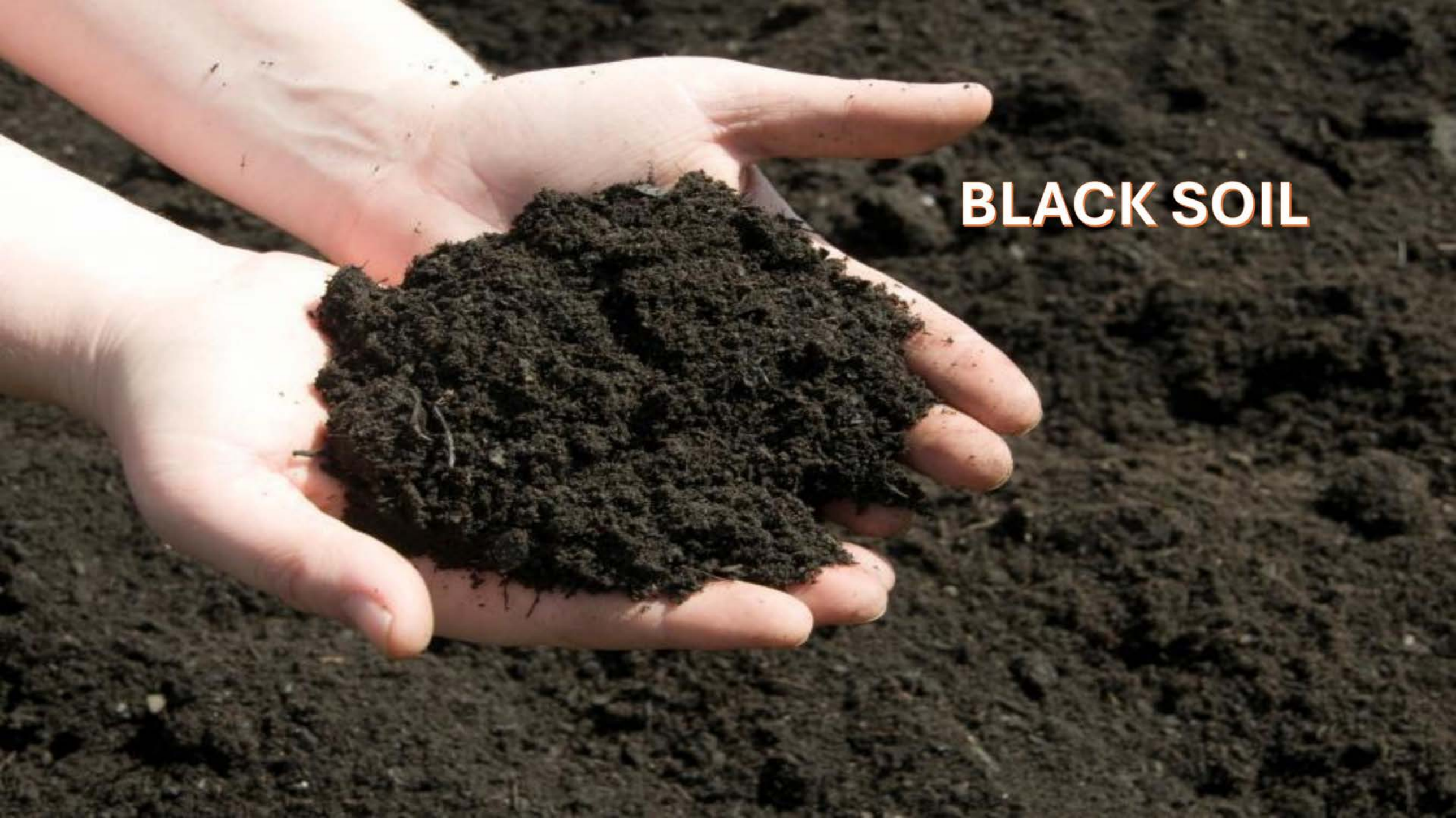
Minerals

- Alluvial soil varies in nature from sandy loam to clay. It is a fertile soil as it is rich in minerals, especially iron, magnesia, alumina, potash and lime but poor in phosphorous, nitrogen and organic matter.
- It is deficient in nitrogen and humus, except the alluvium of the Ganga deltaic region which is rich in humus.
- In the drier areas, it is more alkaline.

1. Alluvial Soil

- **Crops:** Alluvial soil is light and porous and therefore, easily tillable. It is suitable for the growth of a large variety of rabi and kharif crops.
- It is fertile and suitable for cultivation of rice, wheat, sugarcane, cotton, tobacco, gram and oilseeds.
- In the lower Ganga-Brahmaputra Valley it is useful for jute cultivation.


Saturated


BLACK SOIL

1. Which soil type covers the largest geographical area in India?

A) Black Soil

B) Red Soil

C) Alluvial Soil

D) Laterite Soil

3. The older, coarser, and less fertile alluvial soil found in the upper reaches of the Indo-Gangetic plain is known as:

A) Khadar

B) Bhangar

C) Bhabar

D) Tarai

4. Why is Khadar soil more fertile than Bhangar soil?

A) It is rich in iron oxides.

B) It contains a high amount of kankar (lime nodules).

C) It is renewed annually by floods, bringing fresh silt.

D) It has a higher clay content that retains moisture.

Thank You